Final Project Summary and Tech Stack

Generated: 2026-05-16 17:24

HOSPITAL MANAGEMENT SYSTEM FINAL BLUEPRINT

A. FINAL STACK RECOMMENDATION
Frontend:
- React + TypeScript + Vite
- Tailwind CSS
- shadcn/ui
- TanStack Query
- TanStack Table
- Recharts
- SignalR client

Backend:
- ASP.NET Core Web API
- Clean Architecture
- Modular Monolith
- EF Core for main workflows
- Dapper for read-heavy queries
- Hangfire, SignalR, Serilog, FluentValidation

Database:
- SQL Server now
- Oracle-ready design later
- Heavy imaging files kept outside DB in DICOM/PACS + object storage

Reporting:
- QuestPDF for PDFs
- SSRS / FastReport for enterprise reports
- Dashboard analytics in React
- AI narrative reporting later

AI (last phase):
- Python FastAPI
- Azure OpenAI / OpenAI
- AI note assistant
- AI discharge summary
- AI analytics and anomaly detection
- Human review and audit trail required

Deployment:
- Docker from day one
- Kubernetes later for enterprise scale
- Redis, object storage, monitoring, CI/CD

B. HIGH-LEVEL SYSTEM FLOW
Patient registration → Appointment/Visit → Consultation → Orders → Lab/Radiology/Pharmacy → Admission if needed → Floor/room/bed care → Billing → Discharge → Follow-up.

C. KEY DESIGN PRINCIPLES
- Workflow-first design
- Role-based experience
- Auditability
- Modular build strategy
- Object storage / PACS for imaging
- AI as later add-on, not part of MVP

D. PROJECT EXECUTION ROADMAP
Phase 1: Foundation
Phase 2: Registration, appointments, queue
Phase 3: OPD and consultation
Phase 4: Admission, building/floor/room/bed, nursing
Phase 5: Lab, radiology, pharmacy
Phase 6: Billing, reporting, audit
Phase 7: OT, inventory, advanced operations
Phase 8: Deployment hardening and enterprise readiness
Phase 9: AI integration

E. DELIVERY STRATEGY
Use Codex in controlled phases, with each phase generating:
- Domain models
- API endpoints
- UI flows
- Validation
- Tests
- Documentation